

# A uniform entropy bound for fibres of $k\sigma(k)$

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[Code and formal proofs](#)

## Abstract

The number  $f(n)$  of positive integers  $k$  satisfying  $k\sigma(k)=n$  has an all-target upper bound  $\exp((C+\varepsilon)\log n/\log \log n)$ , where  $C=\log(2539/1000)/3+\log(10/9)/2\approx 0.36327036$ . This refines the located  $\log 3/3$  coefficient while leaving the requested zero-constant asymptotic open.

## The exact arithmetic count

Let  $\sigma(k)$  be the sum of positive divisors of  $k$ , and let  $f(n)=\#\{k\geq 1:k\sigma(k)=n\}$ . Since  $k\sigma(k)\geq k$  for positive  $k$ , the solutions form an explicit finite subset of  $\{0,\dots,n\}$ ; the source proves identity with the canonical formal-conjecture count, including its positivity convention.

For every  $\varepsilon>0$ , the result gives a threshold  $N$  such that all  $n\geq N$  satisfy  $f(n)\leq \exp((C+\varepsilon)\log n/\log \log n)$ , with  $C=\log(2539/1000)/3+\log(10/9)/2$ . The numerical comparison is all-target:  $C$  is smaller than  $\log 3/3\approx 0.366204$ . A separately located  $\log 2/2$  bound for odd targets does not subsume a theorem for all  $n$ .

## Powerful cores and an exact pattern bound

Split  $k$  into its powerful core, containing prime exponents at least two, and a coprime squarefree residual part. The squarefree map  $s\mapsto s\sigma(s)$  is injective; the formal argument removes the largest prime and treats the small primes separately. Thus a solution in a fixed target fibre can be encoded by its core exponent pattern without multiplying by an uncontrolled number of squarefree choices.

The equation  $k\sigma(k)=n$  constrains both divisibility and size of that core. The development establishes exact weighted exponent-pattern bounds, with local weights and a cubic estimate at a fixed rational parameter. These finite inequalities yield the  $2539/1000$  constant, rather than introducing an unproved entropy heuristic.

A logarithmic cutoff divides target primes into small and large ranges. Small-prime choices are bounded crudely but contribute a lower-order term; large-prime exponents obey a log-size budget. The proven weighted finite estimate is then converted to the exponential form, and a real asymptotic bridge absorbs lower-order terms into  $\varepsilon$ . Every premise of the intermediate conditional bridge is discharged by the final `SigmaFiberAsymptotic` theorem.

## Mathematical and formal boundaries

Erdős 1060 asks for a little- $o$  exponent, corresponding to arbitrarily small leading constant. A fixed positive  $C$  does not establish that conclusion. The result here is a uniform quantitative partial improvement.

## Prior work and attribution

Powerful-core decomposition, squarefree injectivity and cutoff counting are prior machinery. The refinement is the exact weighted-pattern estimate and its all-target coefficient. Comparison with Kominers and related sources must retain their different target domains and quantitative hypotheses; no best-known assertion follows from a different-domain bound.

The proof uses the local rational inequality, core injection and cutoff argument described above. The related cube-sum theorem shares some of these background modules; this common infrastructure is not an additional independent contribution.

## Formal verification

Lean 4.33.1 checked all ten source modules and the three selected declarations `canonical_count_eventually_upper`, `canonical_count_upper` and `entropyConstant_lt_old`. They use only standard kernel axioms, with no admitted proof, native evaluation or unknown axiom. Their source files and exact audit records are supplied in the companion repository. The zero-leading-constant target remains open.

### Erdos1060.canonical\_count\_eventually\_upper

```
theorem canonical_count_eventually_upper {ε : ℝ} (hε : 0 < ε) :  
  ∀f n : ℕ in atTop,  
  ({k ≤ n | k * ArithmeticFunction.sigma 1 k = n} : ℝ) ≤  
  Real.exp (((Real.log (2539 / 1000)) / 3 + Real.log (10 / 9) / 2) + ε) *  
  Real.log (n : ℝ) / Real.log (Real.log (n : ℝ))
```

Source: `proofs/erdos1060-uniform-partial/src/SigmaFiberAsymptotic.lean:38`.

Source SHA-256: 440f2f6aaa1d5dd04611fe855077352a3b75171d6e1ec823939ede8609e2d45a

### Erdos1060.canonical\_count\_upper

```
theorem canonical_count_upper {ε : ℝ} (hε : 0 < ε) :  
  ∃ N : ℕ, ∀ n : ℕ, N ≤ n →  
  ({k ≤ n | k * ArithmeticFunction.sigma 1 k = n} : ℝ) ≤  
  Real.exp (((Real.log (2539 / 1000)) / 3 + Real.log (10 / 9) / 2) + ε) *  
  Real.log (n : ℝ) / Real.log (Real.log (n : ℝ))
```

Source: `proofs/erdos1060-uniform-partial/src/SigmaFiberAsymptotic.lean:49`.

Source SHA-256: 440f2f6aaa1d5dd04611fe855077352a3b75171d6e1ec823939ede8609e2d45a

**Sources and attribution:** 1. [www.erdosproblems.com / 1060](http://www.erdosproblems.com / 1060); 2. [scottkom.com / Kominers\\_ksigmak.pdf](http://scottkom.com / Kominers_ksigmak.pdf); 3. [google-deepmind/formal-conjectures / 1060.lean](http://google-deepmind/formal-conjectures / 1060.lean); 4. [alejandrozululam-opdp-difficulty-atlas](http://alejandrozululam-opdp-difficulty-atlas); 5. [alejandrozululam-openmath-2026-judging / E02-erdos1060-uniform-partial.txt](http://alejandrozululam-openmath-2026-judging / E02-erdos1060-uniform-partial.txt); Proof source repository.

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